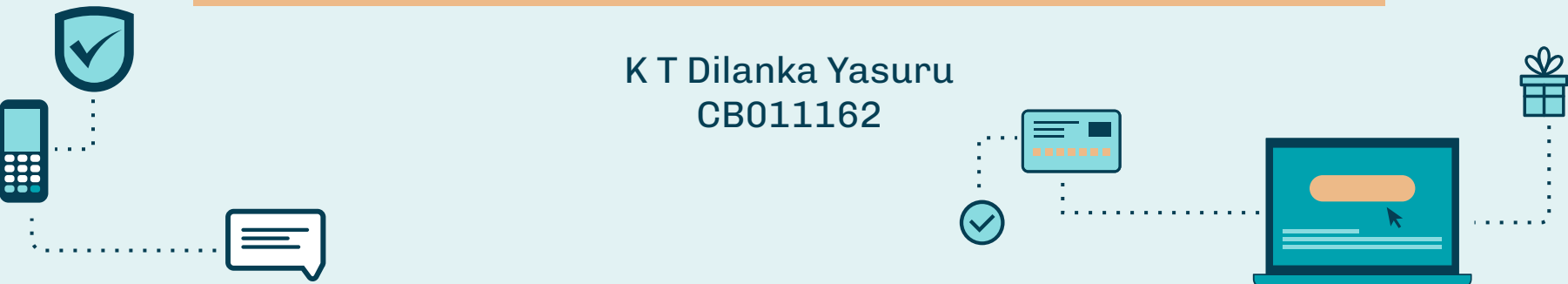


AquaSense AWS Migration

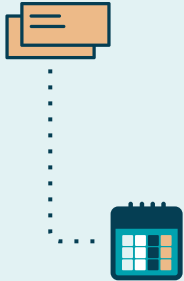
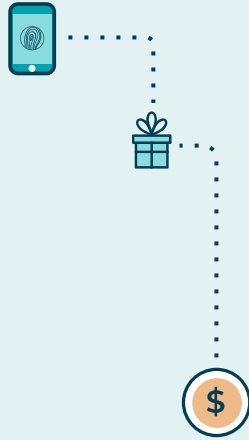
Cloud Architecture and Migration Plan

K T Dilanka Yasuru
CB011162



INTRODUCTION

- AquaSense Smart Utilities (ASU) is a major company providing smart water and energy metering solutions.
- The volume of data from 30,000+ real-time sensors is rapidly increasing.
- The current on-premises infrastructure faces limitations in scalability and advanced analytics capabilities.
- Strategic Goal: Migrate to Amazon Web Services (AWS) to overcome capacity limits, ensure regulatory compliance, and enable predictive insights.



Current Infrastructure Analysis & Cloud Justification

Current State

A legacy on-premises VMware cluster hosting around 180 virtual machines.

Challenges

The monolithic architecture struggles to scale during peak periods and results in expensive, idle assets during off-peak times.

Business Driver

ASU needs to cut infrastructure and operational costs by at least 35%.

Solution

Move to a highly available, event-driven, microservices-based architecture natively on AWS.



Justifying the AWS Cloud Adoption

Scalability

Amazon EC2 Auto Scaling will automatically match compute resources to real-time traffic demand, and AWS Lambda will provide serverless processing for massive telemetry data

Storage & Analytics

Amazon S3 will act as the data lake for historical telemetry, powering AI capabilities.

Resilience

Amazon RDS and Aurora will replace on-premises databases, offering managed high availability and automated failover

Cost Savings

The pay-as-you-go model, combined with AWS Savings Plans, eliminates fixed capital costs.



Cloud Migration Strategy

01. Discovery

Using AWS Application Discovery Service to evaluate 180 VMs and prioritize migration, tracked via AWS Migration Hub.

02. Pilot Migration

Moving a test database safely using AWS Database Migration Service (AWS DMS) to validate the target environment.

03. Approaches

- **Rehost:** Rapid server conversion via AWS MGN.
- **Replatform:** DBs to Aurora/RDS.
- **Refactor:** Monolith to Microservices.

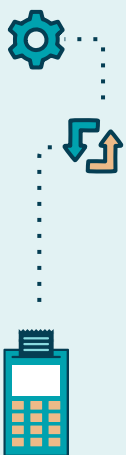
04. Modernization

Utilizing Amazon ECS with Fargate for container orchestration and an event-driven serverless architecture for peak efficiency.

05. Production Rollout

Implementation of Blue/Green deployment strategies with comprehensive CloudWatch monitoring to ensure zero downtime.





Target AWS Architecture Breakdown : (SWP & SMDP)



Customer Web Portal (CWP) – High-Availability User Workloads

- Users access the portal through **Route 53** (DNS routing)
- Traffic flows through **CloudFront** (CDN for fast content delivery)
- Requests are distributed via an **Application Load Balancer (ALB)**
- The application runs on an **ECS Fargate Cluster** with auto-scaling containers
- Data is stored in **Aurora PostgreSQL** with Multi-AZ database failover, ensuring 95% uptime



Smart Meter Data Platform (SMDP) – Event-Driven IoT Ingestion

- **30,000+ smart meters** send data continuously
- Data is ingested through **AWS IoT Core** using the MQTT protocol
- An **SQS Queue** decouples ingestion from processing and is backed by Dead-Letter Queues for error handling
- A **Lambda Event Processor** processes the queued meter data in a serverless, event-driven manner
- Processed data is stored in **DynamoDB (Hot Storage)** for fast access and retrieval



Multi-layered Security & GDPR Governance

Layer 1 : Access & data

- **IAM** — Least Privilege access with Multi-Factor Authentication (MFA)
- **AWS KMS** — Encryption at Rest and in Transit



Layer 2 : Network & VPC

- **Security Groups** — Stateful isolation of resources
- **NACLs** — Stateless subnet-level traffic control



Layer 3: Edge & Perimeter

- **AWS WAF** — Blocks SQL injection and Cross-Site Scripting (XSS) attacks
- **AWS Shield** — DDoS mitigation
- **CloudFront** — Secure content delivery at the edge

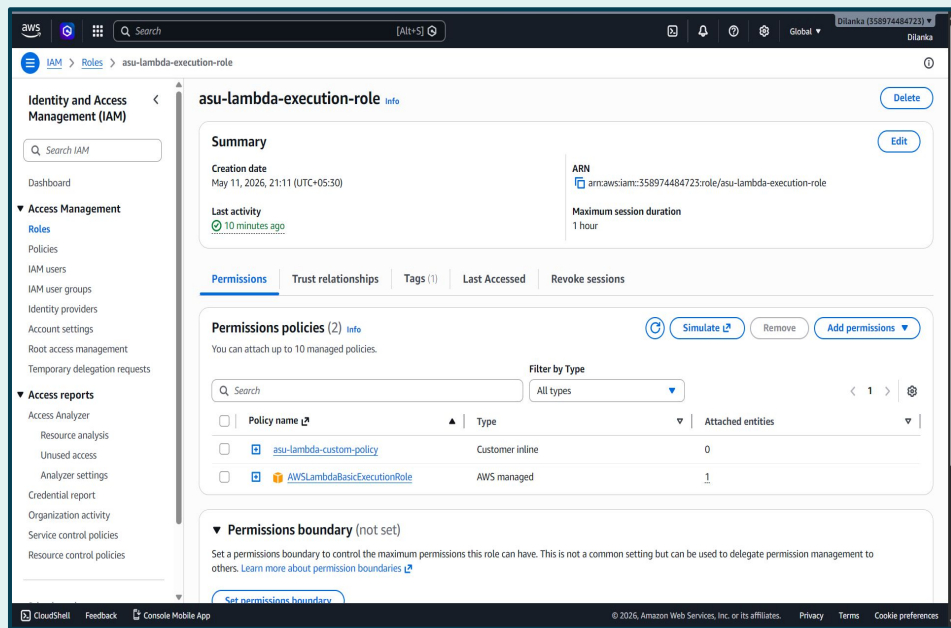
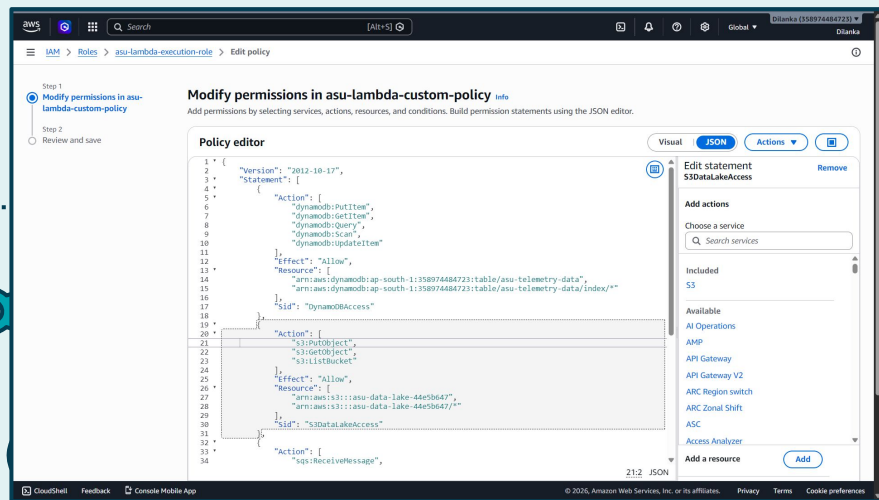
Continuous Governance & GDPR Alignment

- **AWS CloudTrail** — Immutable API audit logs to satisfy GDPR data processing tracking
- **AWS Config** — Continuous posture assessment against security baselines
- **AWS Security Hub** — Centralized security alert management

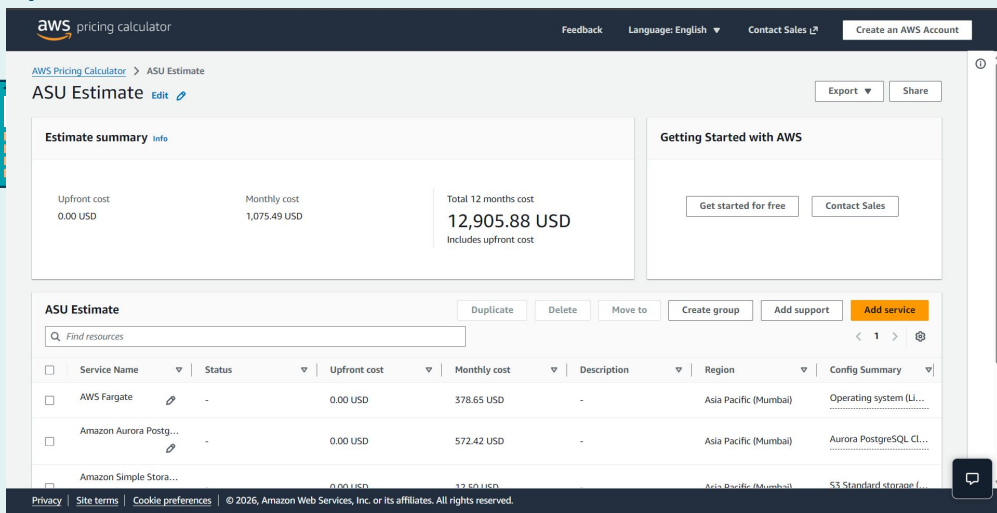


Enforcing Least Privilege Access

- Systems use **IAM roles with temporary credentials** (Assumed Role) — no long-lived access keys
- A **Lambda function** assumes a role and receives temporary credentials to access only what it needs
- All systems are **explicitly denied access** to all resources except those explicitly granted by scoped IAM policies, protecting PII



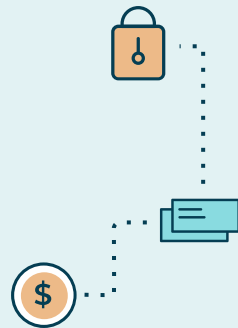
Cost Optimization & Financial Support Strategy



- Estimated Monthly Run Rate: **\$1,075.49**
- Estimated Annual TCO: **\$12,905.88 (Zero Upfront)**
- Top cost: Database (Aurora Multi-AZ) at \$572.42/month

Action Panel — Cost Savings Levers to Exceed 35% Mandate

- **Compute Savings Plans** — Up to 72% off Fargate/Lambda costs
- **S3 Lifecycle Policies** — Automated archiving of cold telemetry data to S3 Glacier
- **AWS Compute Optimizer** — ML-driven right-sizing recommendations



Low-Downtime Data Migration Strategy

1. Initial Full Load:

Bulk extraction from on-prem
MySQL/PostgreSQL



2. Change Data Capture (CDC):

Real-time transaction replication to
Aurora

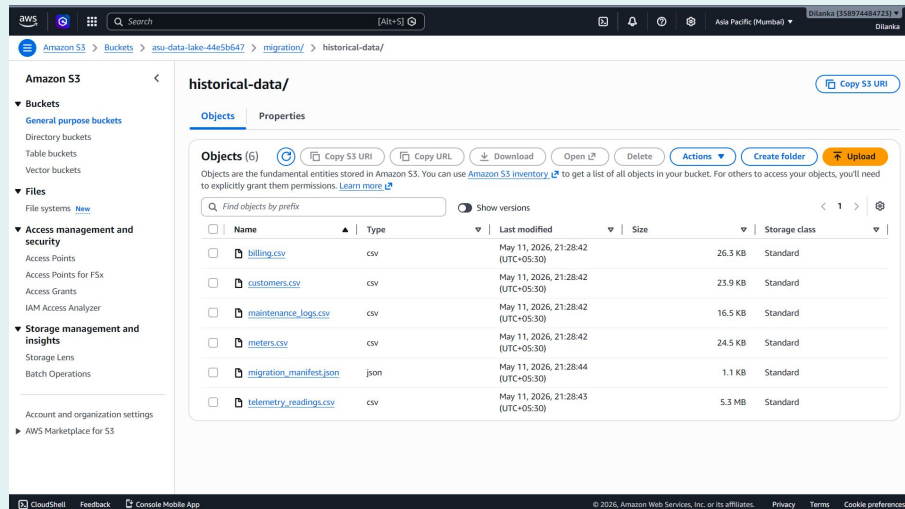


3. Cutover:

Near-zero downtime
switchover to AWS

Data Validation

AWS Glue performs automated checksums
and row counts to ensure data integrity



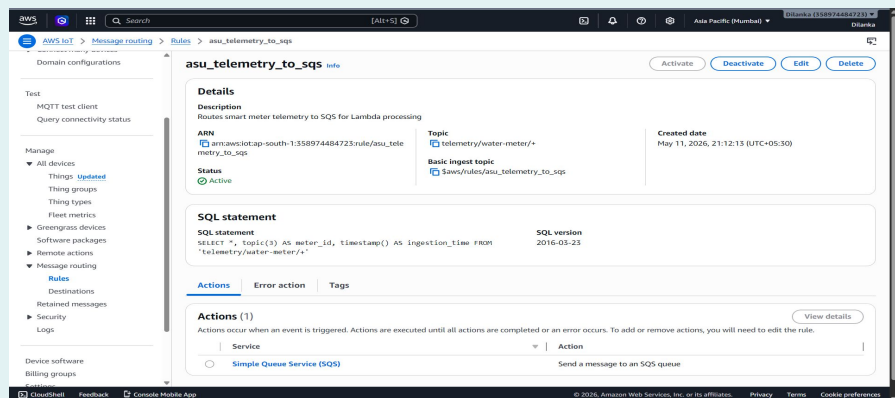
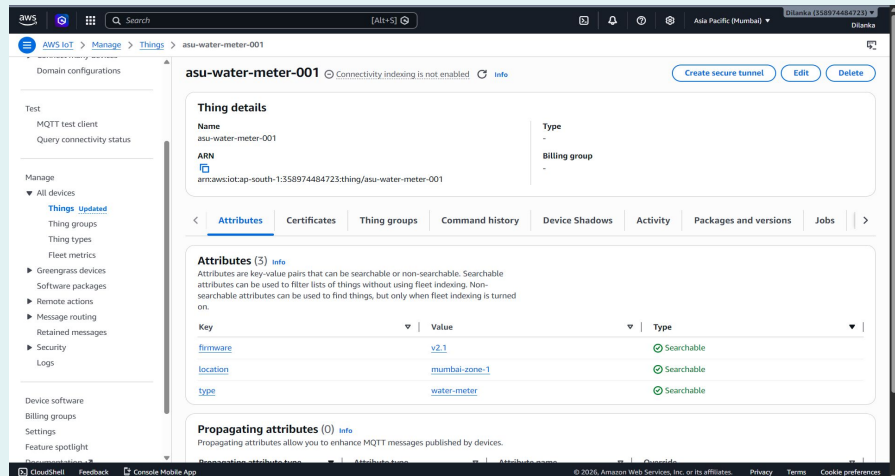
Hybrid Connectivity & Edge Integration

Corporate Network

On-prem VMware connects to AWS via Direct Connect (primary) and Site-to-Site VPN (failover) through a Virtual Private Gateway into a Multi-AZ VPC

Edge IoT Network

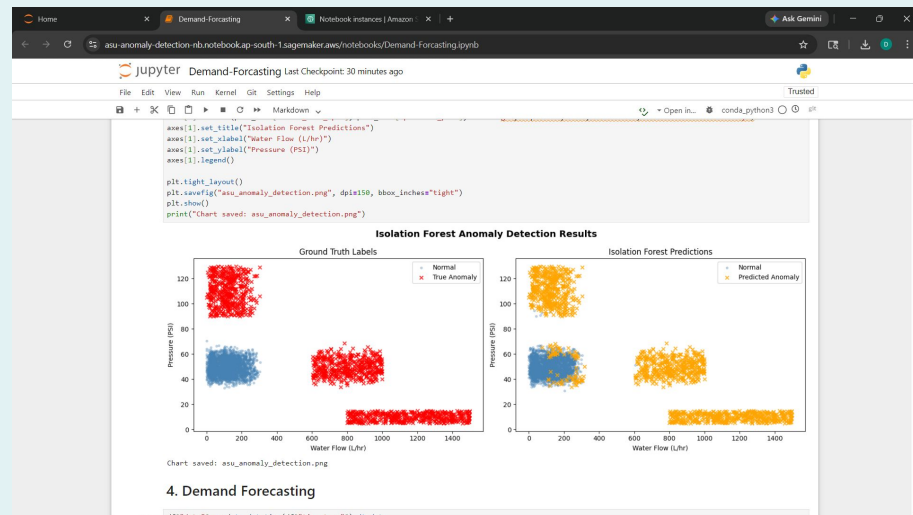
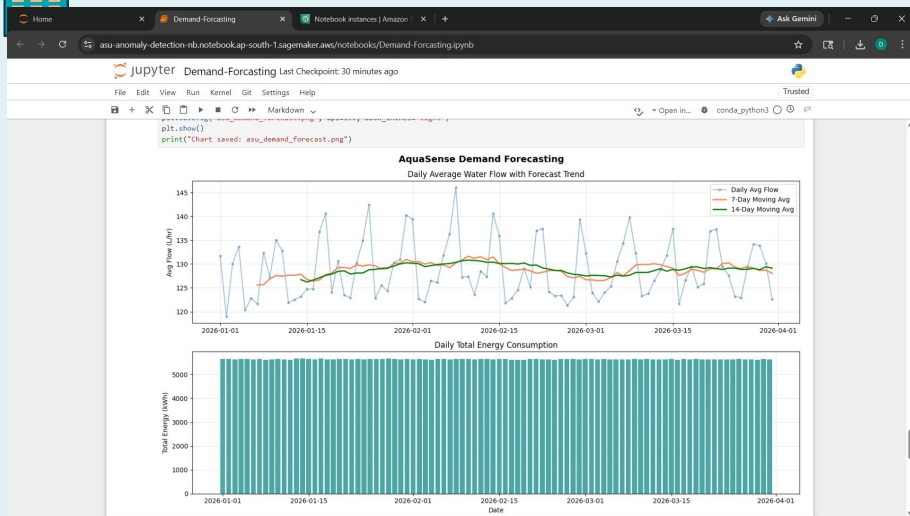
30,000+ smart meters connect via AWS IoT Greengrass Edge Gateway using MQTT protocol to AWS IoT Core



AI-Driven Innovation & Predictive Analytics

Pipeline Flow : IoT Rules → Kinesis Firehose → S3 Data Lake → Amazon SageMaker

Amazon QuickSight



Implementation Proof I: Core Infrastructure

Control Panel

VPC dashboard < Subnets (1/7) info

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-075910e9f6d84432f	Available	vpc-056f69d2aa4d92197	Off	172.31.0.0/20
-	subnet-0a4503c341cc74c4f	Available	vpc-056f69d2aa4d92197	Off	172.31.32.0/20
☒ asw-public-subnet-3	subnet-0efa49e0f1262cdc4	Available	vpc-08f5ca0a4729d3b50 asw-...	Off	10.0.1.0/24
-	subnet-0f53f25d4102a3cf7	Available	vpc-08f5ca0a4729d3b50 asw-...	Off	10.0.3.0/24
-	subnet-0f0a8f215ab804b32	Available	vpc-08f5ca0a4729d3b50 asw-...	Off	10.0.4.0/24
-	subnet-02370b6cde516a7de	Available	vpc-056f69d2aa4d92197	Off	172.31.16.0/20
-	asw-public-subnet-2	Available	vpc-08f5ca0a4729d3b50 asw-...	Off	10.0.2.0/24

subnets-0efa49e0f1262cdc4 / asw-public-subnet-1

Details | Flow logs | Route table | Network ACL | CIDR reservations | Sharing | Tags

Subnet ID	Subnet ARN	State	Block Public Access
subnet-0efa49e0f1262cdc4	arn:aws:ec2:ap-south-1:35897448472:3:subnet/subnet-0efa49e0f1262cdc4	Available	Off

IPv4 CIDR	Available IPv4 addresses	IPv6 CIDR	IPv6 CIDR association ID
10.0.1.0/24	251	-	-

Availability Zone: ap-south-1a

VPC dashboard < Route tables (1/4) info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
asw-private-rt	rtb-0158a37497bd05495	2 subnets	-	No	vpc-08f5ca0a4729d3b50 asw-...
☒ asw-public-rt	rtb-0e56d4f7d456dd591	2 subnets	-	No	vpc-08f5ca0a4729d3b50 asw-...
-	rtb-05ce591d44f6a51b	2 subnets	-	Yes	vpc-08f5ca0a4729d3b50 asw-...
-	rtb-047ee3307e632e99a1	-	-	Yes	vpc-056f69d2aa4d92197

rtb-0e56d4f7d456dd591 / asw-public-rt

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0c12f6f68a08b852c	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Foundation: Multi-AZ VPC subnet configuration

Foundation : Internet Gateway Route Table Association

Implementation Proof II: Compute & Edge

Control Panel

The screenshot displays the AWS Lambda console for the 'asu-event-processor' function. The 'Function overview' section shows the function's ARN, description, and last modified time. The 'Monitor' tab is active, showing CloudWatch metrics for invocations, duration, and error count.

Function overview

- Function ARN: `arn:aws:lambda:ap-south-1:358974484723:function:asu-event-processor`
- Description: Processes IoT telemetry data from SQS, writes to DynamoDB and S3
- Last modified: 42 minutes ago

Monitor

Filter metrics by: Function View CloudWatch logs

CloudWatch metrics

Lambda sends runtime metrics for your functions to Amazon CloudWatch. The metrics shown are an aggregate view of all function runtime activity. To view metrics for the unqualified or SLATEST resource, choose **Filter metrics by**. To view metrics for a specific function version or alias, choose **Aliases** or **Versions**, select the alias or version, and then choose **Monitor**.

Metric	Count	Duration (Milliseconds)	Error count and success rate
Invocations	48	792.46	1

Compute: Serverless Event Processor with SQS Trigger

The screenshot displays the AWS API Gateway console for the 'asu-customer-web-portal-api' deployment. The 'Resources' section shows the API's structure, including the '/meters' resource. The 'Method request settings' section shows the configuration for the 'GET' method.

API Gateway

Resources - asu-customer-web-portal-api (p5g3yellymb)

Resources

- /health GET
- /meters GET
- /meters/{meter_id} GET
- /status GET

/meters - GET - Method execution

ARN: `arn:aws:execute-api:ap-south-1:358974484723:p5g3yellymb/GET/meters` Resource ID: `j5mjbni`

Method request settings

Authorization: NONE API key required: False

Request validator: None SDK operation name: Generated based on method and path

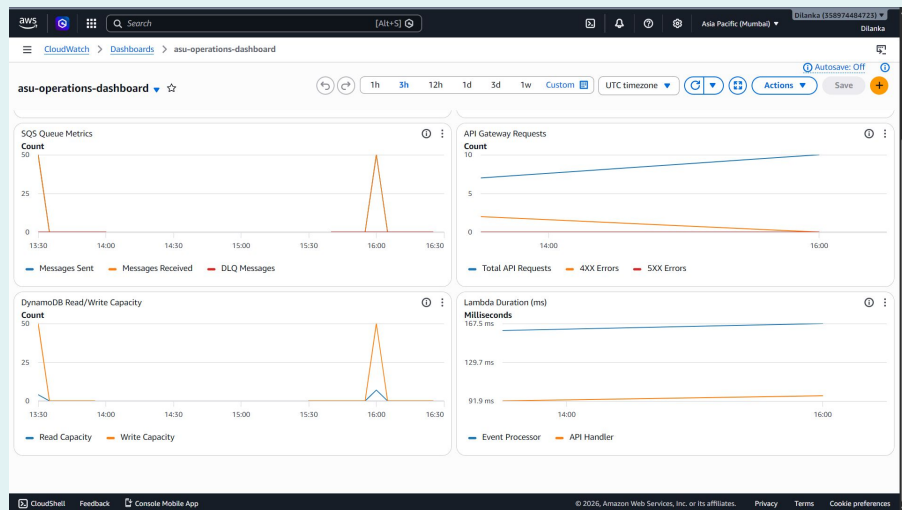
API: Customer Web Portal REST Endpoint Deployment



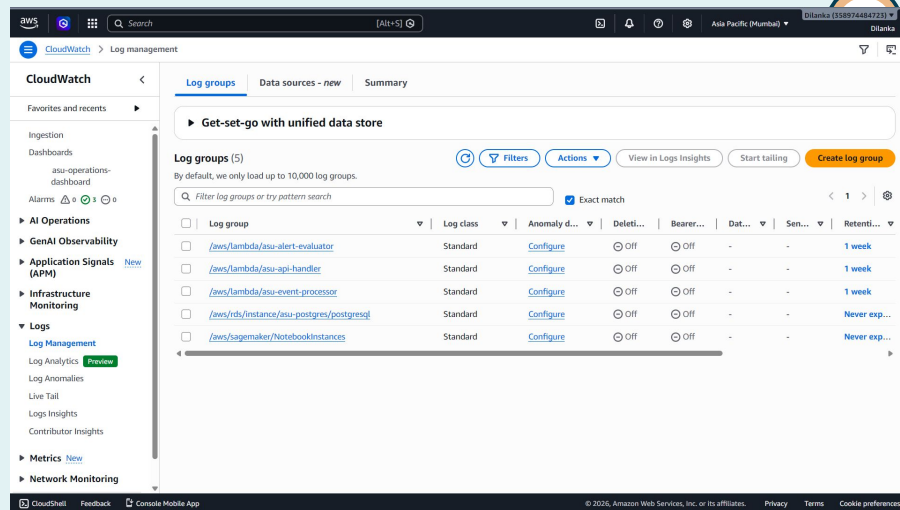
Implementation Proof III: Observability



Control Panel



Monitoring: Cloud watch operations dashboard



Governance: Active CloudWatch Application Log Groups



Conclusion

This AWS architecture addresses every core business requirement through a secure, scalable, and cost-efficient cloud foundation. With Multi-AZ deployments ensuring 95% availability, an event-driven IoT pipeline processing 30,000+ smart meters in real-time, and serverless design cutting costs by over 35%, the solution is both powerful and economical. Integrated AI capabilities through SageMaker further enable predictive demand forecasting, making this a future-ready platform for intelligent utility management.



THANK YOU!

